Explaining Monthly Variation in Auckland Bus Punctuality

Project summary | Master of Business Informatics capstone (MBI908), Yoobee Colleges, November 2026 | Eddy (Jiada) Jiang

THE QUESTION

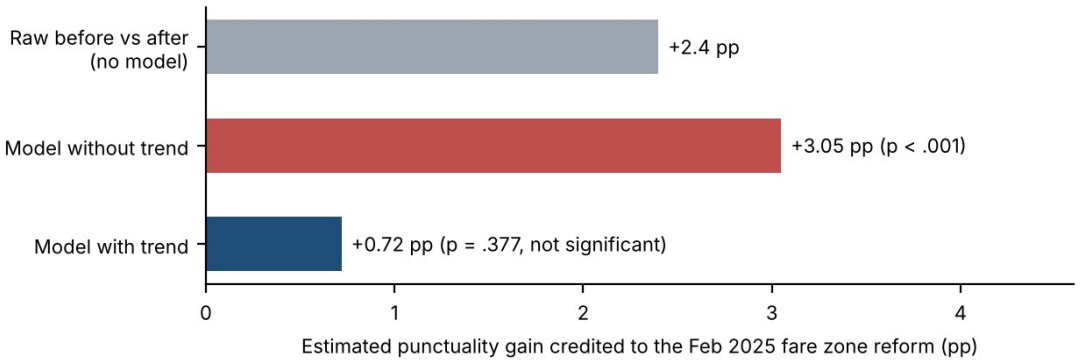
Auckland Transport publishes one monthly bus punctuality figure. It averaged 89.7% from July 2023 to June 2026, sets the bonuses and abatements paid to bus operators, and is the headline the Council and public see, yet it comes with no explanation of why it moves. The study asks how far three dated events (the end of the driver shortage, the Rail Network Rebuild closures and the February 2025 fare zone reform), together with seasonal and holiday factors, explain that movement.

DATA AND METHOD

36 months of public AT workbooks (punctuality, boardings and Bus Performance Reports) plus a panel of 6,601 route-months. An interrupted time series regression with a pre-specified trend, a seasonal SARIMA forecasting baseline, route fixed effects and a difference-in-differences test, with the minimum detectable effect reported for every null result. Raw files are hashed and the full analysis can be reproduced from public data.

FOUR FINDINGS

1. **Punctuality moves with the calendar and how busy buses are.** Each extra boarding per service hour is linked to 0.69 pp lower punctuality; ridership and the trend alone explain about 75% of monthly variation.
2. **No named event is detectable, and the apparent fare effect is an artefact.** Without the trend the fare reform appears worth +3.05 pp; with the trend it falls to +0.72 pp and is not significant (chart below).
3. **Event information adds no forecasting value.** A seasonal model using only past punctuality forecast more accurately (RMSE 0.98 pp) than the full regression (1.37 pp).
4. **The headline hides where punctuality is poor.** School services average 73.2% and South Auckland is lowest among scheduled routes (86.6%). A definition change in July 2023 (96.9% old vs 87.9% new) is larger than three years of improvement.



The improvement after February 2025 is real, but the data cannot attribute it to the reform rather than to improvement already under way.

SIX RECOMMENDATIONS TO AT

- **Trend-controlled attribution:** test any claimed intervention effect against the trend and calendar, and publish its confidence interval.
- **Ridership reporting:** report boardings per service hour beside punctuality to target bus priority and faster boarding.
- **Seasonal expectation band:** call a month "changed" only when it falls outside its seasonal range.
- **Area and service-level reporting:** publish results by area and report School services separately.
- **Release metadata:** state the definition, route population and excluded days with every release.
- **Separate contractual and customer measures:** add interim stops at key interchanges, as the Auditor-General suggested.

WHY IT MATTERS FOR REVIEWING BUSINESS CASES

The fare result shows how a real improvement can be credited to the wrong intervention when the underlying trend is left out. The same test applies when reviewing the benefits claimed in a business case: is the benefit caused by the project, or would it have happened anyway?